

PROJECT DESIGN PHASE

Solution Architecture

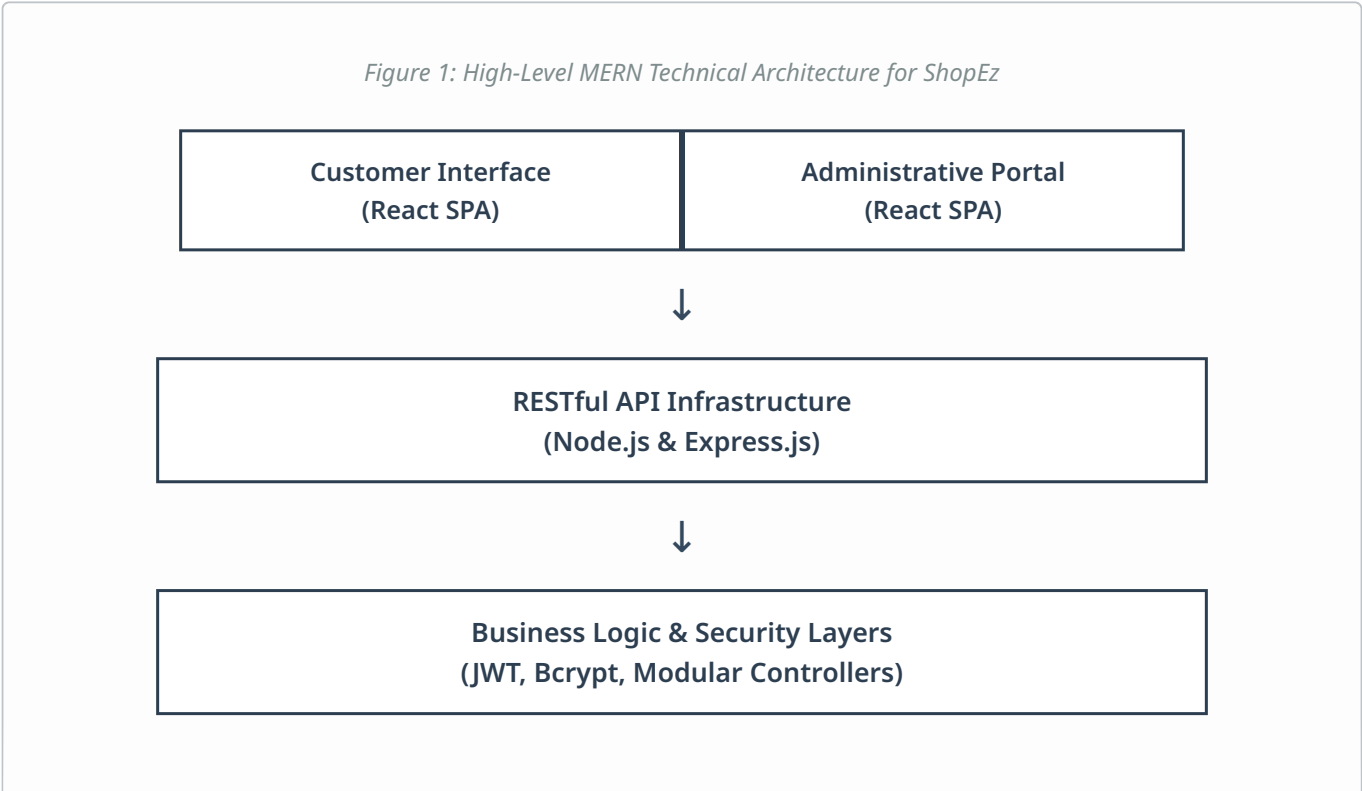
Date	18 June 2026
Team ID	
Project Name	ShopEz (Full Stack MERN-Based E-Commerce)
Maximum Marks	4 Marks

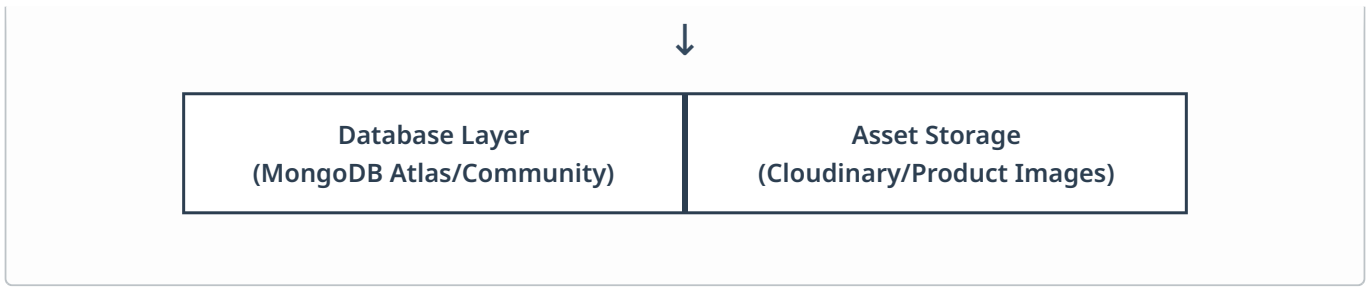
Solution Architecture Overview:

Solution architecture serves as a foundational engineering framework that aligns high-level business goals with technical execution. For the ShopEz platform, the system is architected to facilitate scalable product catalog management and robust transaction processing by utilizing the modern MERN technology stack.

- Utilize the MERN ecosystem to address core e-retail navigation and administrative complexities.
- Implement a strongly typed schema for Users, Inventory, and Transactions using MongoDB.
- Guarantee protected API interactions through Express middleware and JWT-based authorization.
- Ensure a low-latency user experience by deploying a React-powered Single Page Application (SPA).

ShopEz - Solution Architecture Diagram:





Detailed Component Analysis:

Presentation Layer (React): Comprises two separate frontend modules designed to offer seamless browsing for shoppers and specialized management (CRUD) interfaces for store staff via a component-driven architecture.

Server & API Gateway (Node/Express): Orchestrates encrypted HTTP communication, validates incoming payloads, and manages routing logic to ensure high system availability and efficient asynchronous data streaming.

Persistence & Storage (MongoDB): Leverages flexible NoSQL collections for storing Customers, Stock, and Orders, integrated with cloud-based storage solutions for high-resolution product imagery.

System Data Flow: Front-end interactions initiate API requests; the backend validates user identity via JSON Web Tokens, executes queries against MongoDB, and transmits JSON payloads for real-time dynamic rendering in React.